

VARAD GANDHI

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EDUCATION

University of Illinois Urbana-Champaign, Gies College of Business

Champaign, IL

Master of Science in Finance (STEM designated program), **GPA: 3.7/4**

August 2024 – August 2025

- **Relevant Coursework:** Accounting, Investments, Advanced Python, Economics, Managing Market Risks, Quant Finance, Fixed Income Portfolios, Applied Financial Analysis, Big Data Analytics in Finance
- **Extra-Curriculars:** Teaching Assistant/Growth Group Facilitator for BUS 401

NMIMS University, MPSTME

Mumbai, India

MBA in Technology Management, **GPA: 3.8/4**

July 2019 – August 2024

- **Relevant Coursework:** Business Visualization, Programming for Analytics, Project Management, Financial Analysis, Financial Risk Management, Financial Portfolio Management, Spreadsheet Modelling
- Winner – Student Research Competition for the research on “Game theoretical analysis in Hospitality Industry”

NMIMS University, MPSTME

Mumbai, India

Bachelor of Technology in Information Technology (Honors), **GPA: 3.8/4**

July 2019 – August 2024

- Honors with a specialization in **AI/ML** in collaboration with IBM Innovation center for excellence
- **Relevant Coursework:** Python, Advanced Database Management, Cloud Computing, AI/ML, OOPS, Data Warehousing

EXPERIENCE

IDX Insights

San Ramon, CA

Quant Projects Analyst - Contract

September 2025 – Present

- Engineered GPT-based AI agents for market risk forecasting and for company-level financial analysis, generating a full-fledged equity research report for valuing company performance and debt, reducing manual task by **3+ hours**
- Built a high-frequency BTC forecast engine for the Bitcoin ETF Strategic Fund (BTDIX) using 1-minute CoinAPI data, engineering 40+ features and testing multiple ensembles, selecting a Random Forest that improved 15-min directional AUC by **12%** via walk-forward validation

IDX

Champaign, IL

Intern, Quantitative Research and Technology

June 2025 – August 2025

- Researched and developed proprietary S&P 500 risk-scoring models using Random Forest and regression, supporting systematic portfolio risk classification and strategy rebalancing based on market benchmarks reducing tail risk by **12%**
- Performed scenario analysis in Python for different benchmarks to create a proprietary risk matrix for investments

Nomura Structured Finance Services Limited

Mumbai, India

Intern, Global Markets Division – Electronic Trading Client Services

May 2023 – September 2023

- Built real-time reconciliation frameworks (Python + Alteryx) for trade data from Bloomberg & Tradeweb, enhancing data integrity, reducing operational and pricing risk by **60%**, and improving oversight across FX and Fixed Income flows
- Built a scalable SQL-based MIS with dimensional modeling to manage **10,000+** client trading records, replacing legacy VBA and reducing manual effort by **2+** hours daily and enabling faster BI insights
- Developed and automated a real-time ETL pipeline (Python APIs + Alteryx) to extract/process trade data from Bloomberg & Tradeweb, applying data auditing, transformation, and validation to deliver BI-ready datasets
- Engineered an automated data pipeline using Power BI to process and visualize daily trading data for **10,000+** clients, generating actionable insights into cross-selling opportunities across FX and Fixed Income products

IBM PHEME Soft Pvt Ltd

Mumbai, India

Python Web Developer Intern

May 2022 – June 2022

- Engineered an end-to-end application for the orientation of new hires using Python Django and SQL database technologies
- Developed APIs for seamless information transfer of client credentials ensuring storage integrity saving manual entry time

SKILLS

Technical - Python, SAS, R programming, SQL, VBA, Django and Spark, Alteryx Analytics, Power BI, Tableau, Excel, Risk Management, Machine Learning, Financial Analysis, A/B testing, Back testing, Stress Testing, Financial Modelling

Interpersonal Skills - SOP Documentation, Adaptability, Stakeholder Communication, Leadership, Team collaboration

CERTIFICATIONS

- Google Data Analytics Professional Certification November 2023
- Financial Markets Coursera October 2022

LEADERSHIP

BUS 401, Gies College of Business

Champaign, IL

Teaching Assistant/ Growth Group Facilitator

January 2025 – May

2025

- Leading 20-minute small growth group meetings with 10-20 students per group
- Facilitation of “Growth Groups” to provide personalized, interactive discussions on course concepts and mutual support

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- Completing course grading and developing guides for Growth Group discussion
- Conducted three weekly consecutive Growth Group sessions for BUS 401 students, totaling 80 minutes with short breaks

FINDROME Committee, NMIMS University

Team Director & Head of Finance

Mumbai, Maharashtra

July 2023 – April 2024

- Directed a team of 80+ individuals for ideating, crafting and executing student-led initiatives of 'Market Analysis' and 'Financial Insight' sessions. Hosted the flagship 'Panel Discussion' event on Sustainability and Green Financing
- Led the Capital Research Community for developing and publishing equity valuation and market analysis reports

PROJECTS

Quantification of the effects of economic indicators on the performance of US stock market indices

- Utilized Python and the Yahoo Finance API to extract and process time-series data for major U.S. stock indices
- Applied Multiple Linear Regression and XGBoost to model economic indicators' (GDP, inflation, unemployment) impact on market performance, visualizing insights via heatmaps and correlation matrices to support data-driven forecasting

Cross-Platform A/B Advertising Performance Analysis (Facebook vs AdWords)

- Performed comparative analysis using Python and Power BI, evaluating CTR, conversion, and cost efficiency across Facebook ads campaign and Google AdWords from 2021 to 2024
- Identified Facebook as 2.3x more effective with 37% lower acquisition cost, supporting optimized media spend decisions

Financial Analysis of Textile Sector using Excel and Python

- Leveraged Python and Yahoo Finance API for in-depth analysis of daily stock price movements and trends
- Conducted Ratio Analysis by analyzing the financial statements to evaluate financial health and performance

Investment Proposal & Financial Modeling – Smart Supply Chain Venture

- Developed integrated financial and valuation models (DCF, NPV, IRR, WACC) and built pro-forma financial statements projecting revenue growth from ₹1.6Cr to ₹4.7Cr to assess venture scalability and capital efficiency
- Delivered investment feasibility analysis with NPV ₹78.6M, IRR 38%, and 2.9-year payback, providing data-driven insights for investor decision-making and optimizing capital budgeting and return projections

Accounting Operations and Financial Reporting for a Retail Business

- Executed end-to-end accounting (journals, ledgers, trial balance) for a clothing retailer, managing ₹1 Cr+ in transactions
- Prepared P&L and Balance Sheet, highlighting cost drivers and reporting a net profit of ₹20.05 Lakh

Portfolio Analysis of the NIFTY REALTY Index

- Performed a 33-month quantitative evaluation of NIFTY Realty constituents using CAPM, Markowitz Efficient Frontier, APT, and Sharpe Single-Index models to assess systematic risk, diversification benefits, and portfolio optimization efficiency
- Constructed an efficient frontier and minimum-variance portfolio, achieving a 71% annualized return at 5.6% risk, and applied multi-factor CAPM/APT valuation to identify mispriced securities and optimize risk-adjusted alpha generation

Multi-Asset Derivative Pricing and Risk Modeling application

- Built a Monte Carlo-based Streamlit app to price call options on AAPL, MSFT, and EUR/USD using 100k simulated paths
- Computed portfolio VaR (95%: 27.4%, 99%: 32.4%) and CVaR (95%: 35.6%, 99%: 36.2%), visualizing return and risk

Google Play Store Rating Prediction & Trend Discovery

- Applied regression, clustering, and discriminant analysis on 7,723 apps to identify Reviews, Installs, Category, and Content Rating as key predictors, achieving 78% classification accuracy and revealing price variance ($p = 0.041$) impacting revenue
- Delivered ROI-focused insights linking >9.9M installs and >380K reviews to high-rating tiers, enabling data-driven pricing strategies and portfolio optimization for improved financial performance

Intraday Alpha Modeling & Back testing with Alternative Data

- Developed a walk-forward XGBoost model for intraday equity price prediction, boosting accuracy by 15% using market microstructure and sentiment data of multiple US Indices such as S&P500
- Optimized position sizing, hyperparameters, and transaction costs, achieving 20% higher risk-adjusted returns

LLM-Driven Bollinger Band Trading Framework for S&P ETF Return Optimization

- Engineered and benchmarked AI-driven Bollinger Band trading systems using advanced LLMs (Llama, OpenAI GPT, Claude, and Sonar) to auto-generate, test, and refine trading logic, achieving +140% cumulative returns with a maximum drawdown of 34% over live-simulated S&P 500 ETF data
- Implemented real-time signal generation, position sizing, and risk controls via Python (pandas-ta, NumPy, and asyncio), optimizing code latency and execution speed for high-frequency decision loops

Stress Testing for Financial Portfolios

- Designed and executed macro stress-testing framework on a \$100M multi-asset portfolio, replicating systemic shocks (2008 Financial Crisis, COVID-19 Crash) and quantifying VaR/CVaR at 95%–99% confidence levels to measure tail-risk exposure and capital adequacy, revealing potential losses up to \$25M
- Leveraged XGBoost, PCA, and factor decomposition to isolate key macroeconomic and volatility drivers of portfolio drawdowns, reducing systematic noise by 20% and enhancing model explainability and scenario sensitivity

Structuring of a Leveraged Buyout Deal for Walgreens

- Built a fully integrated LBO model for a \$15 billion Walgreens acquisition, designing a multi-tier capital structure (senior term loans, mezzanine debt, revolver facility) and linking income statement, balance sheet, and cash flow forecasts to evaluate transaction feasibility and investor IRR targets

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- Performed sensitivity and scenario analyses on leverage ratios, exit multiples, and interest coverage to assess capital efficiency, credit risk, and equity returns, optimizing the structure to achieve a 22% target IRR under base-case assumptions

Development of automated reconciliations for risk control in the trading venue setups for trade executions

- Utilizing a combination of Alteryx with embedded Python codes for identifying the points of client information mismatch on the internal Nomura system and the external multi-dealer platforms (Bloomberg & Tradeweb)
- Improved efficiency by 70%, eliminating manual work of reconciling data and applying additional analytics techniques for improving the performance of the Nomura FX platform on handling client and product data

POST IPO Analysis report on DCX Systems

- Conducted deep research and constructed a report on the IPO of DCX Systems in India in 2022, covering a deep analysis of the management structure, the business segment and competition, litigation matters of the company, the capital structure and the fundamental ratio and financial analysis of the company
- In addition to the fundamental analysis, a quantitative post-listing analysis of the company also conducted on multiple technical indicators, justifying the 26.72% annual return of the company in terms of the stock price

Credit Card fraud detection system

- As a part of the IBM Honors program, we constructed a code for credit card fraud detection using multiple algorithms such as Decision Tree, Logistic Regression and SVM to predict credit card fraud/payment default
- Confirmed the performance of the model using confusion matrix, precision recall system as the testing parameters with 94% accuracy on the test data

Larsen & Toubro Finance Equity research report

- Published an extensive and comprehensive study on L&T, incorporating a detailed industry and company parameter analysis alongside a rigorous evaluation of the core financial statements of the past 3 years
- Integrated insights from management discussions, offering a nuanced understanding of the company's current performance & outlook

Enhancing Portfolio Performance with Advanced Options Strategies

- Designed and simulated an options portfolio on Indian equities (HUL, ITC, Infosys, Tata Consumer), applying fundamental & technical analysis with Strip/Strap hedging strategies
- Allocated ₹10L across positions, achieving 18% portfolio return enhancement, mitigating downside risk in volatile markets

Equity Valuation Project – Applied Materials vs. LAM Research

- Built full DCF models and normalized income statements (2023–2024) to evaluate valuation differences
- Analyzed industry trends, margins, ROIC, and market caps (\$130B vs. \$120B) to explain a ~\$10B value gap along with an in-depth industry analysis for predicting future growth trends

Mortgage-Backed Securities Prepayment & Pricing Framework – Rosetree Capital Management

- Stress-tested a \$100M MBS pass-through under ± 100 bps shocks and varying PSA speeds, analyzing prepayment and pricing sensitivity and built an MBS valuation tool using PSA and incentive-adjusted CPR, quantifying prepayment risk, convexity, and cash flow dynamics

Alternative data factor modelling for quantifying its effect on BTC and ETH returns

- Utilized multiple machine learning models to quantify the effect of alternate data factors such as social and news sentiment, economic indicators, On-chain metrics on the returns of Bitcoin and Ethereum
- In addition, we developed a multi-agent system for generating signals based on the analysis of these alternative data factors and back testing for maximizing returns up to 6% with a drawdown of 45% for a volatile asset class like crypto

Machine Learning Pipeline for Oil Profit Forecasting & Strategic Optimization

- Designed and implemented an end-to-end ML pipeline (data cleaning, feature engineering, model training, and evaluation) using Linear/Lasso Regression, Random Forest, and Decision Tree models, achieving R^2 of 0.995 and RMSE of \$4.05B
- Identified Brent crude price dominance (42% variance) and optimal production range (29K–33K bbl/day) as key profit drivers and proposed dynamic hedging

PUBLICATIONS

6th ICRTAC 2023, IEEE

December 2023

Title - *The development of a self-help respiratory ailment mobile application*

Designed a user-friendly app leveraging advanced ML algorithms including MLR, SVM, Random Forest (Regressor, Classifier), and the Decision Tree Classifier to predict COVID variants. Developed custom code to evaluate and identify the best performing model, enabling accurate diagnosis and integrating AI-driven insights with user-friendly healthcare support

ADDITIONAL INFORMATION AND ACHIEVEMENTS

Languages: Marathi (native), English (fluent), Hindi (fluent), Spanish (elementary)

Achievements: Winner at the **Student Research Competition** held at NMIMS University in 2024; Winner of the **Final Year Capstone Project Competition** at NMIMS University in 2023 for development of ML based Respiratory Health Application; Best Research Paper award at **6th ICRTAC 2023 – IEEE Conference** at VIT Chennai; Declared as a **Meritorious Student** for the Batch of 2024 at MPSTME, NMIMS University